

Assignment - 1

Q.1 Explain issues in designing distributed system

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1) Heterogeneity

⇒ The Internet enables users to access services and run applications over a heterogeneous collection of computers and networks.

- Heterogeneity applies to all of the following:-

- i) Hardware devices such as computers, tablets, mobile phones, etc.
- ii) Operating systems such as Windows, Unix, MAC, Linux, etc.
- iii) Programming languages such as Java, C/C++, Python, PHP, etc.
- iv) Different roles of software developer's, designers, system managers.

2) Transparency

⇒ Transparency is the main goal of a distributed system is to hide the actuality of physical distribution of processes and resources across several computers.

- Following types of transparency is present in distributed system:-

- i) Location : Hide where a resource is located.
- ii) Migration : Hide that a resource may move to another location.
- iii) Access : Hide differences in data representation and how a resource is accessed.

3) Scalability

⇒ As the number of users or requests increases, the system must scale accordingly without performance degradation.

- Ensuring performance across geographically dispersed locations.

4) Reliability

⇒ Using duplicate components to take over in case of failures.

- Automatically switching to a standby system when the primary system fails.

5) Security

⇒ Many of the information resources that are made available and maintained in distributed systems have a high intrinsic value to their users.

- Security for information resources has three components:-

i) Confidentiality: protection against disclosure to unauthorized individuals.

ii) Integrity: protection against alterations or corruptions.

iii) Availability: protection against interference with the means to access the resources.

Q.2 Differentiate between NOS, DOS and Middleware in the design of distributed system.

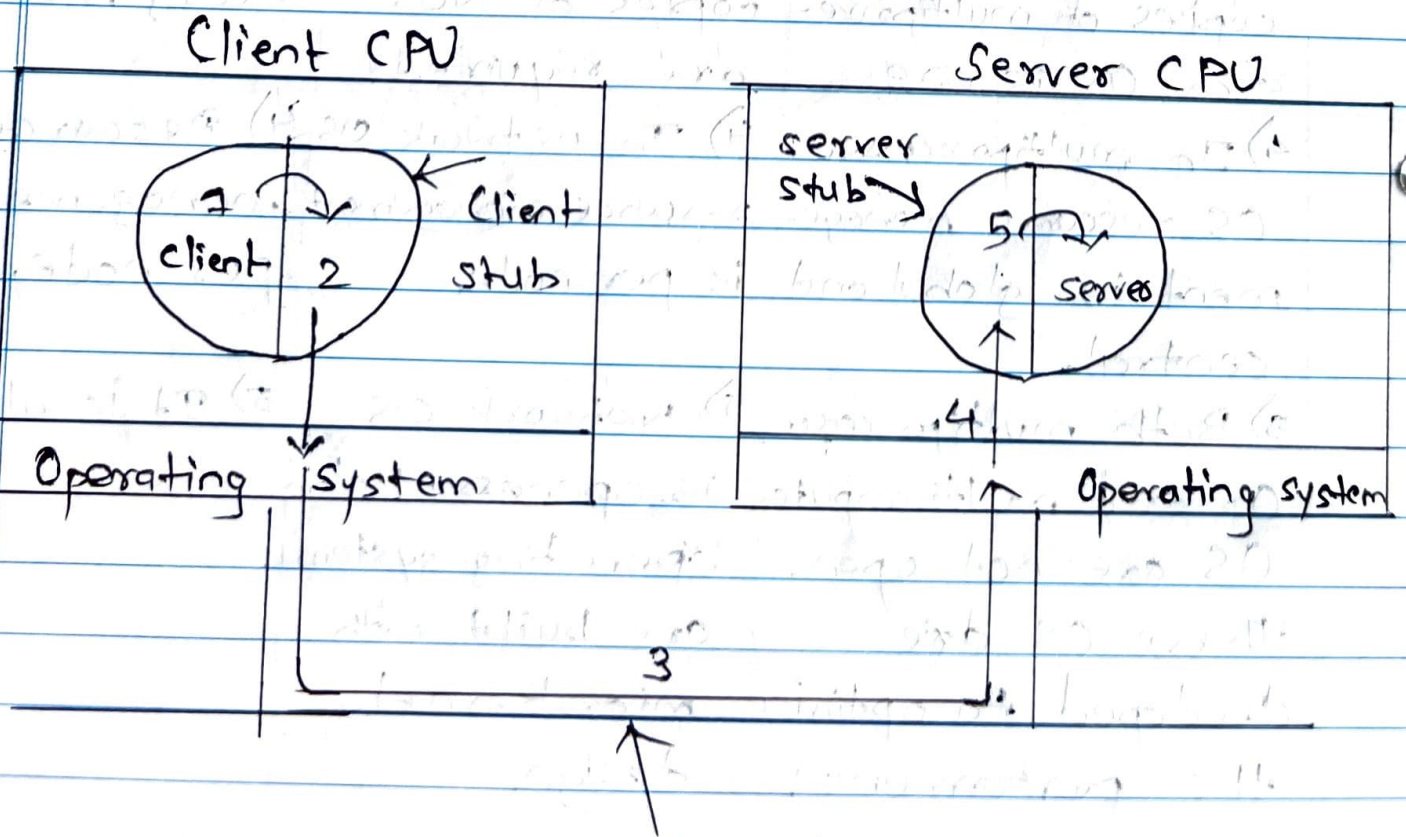
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Distributed OS (DOS)	Network OS (NOS)	Middleware
1) Degree of transparency is very high for multicomputer OS. 2) Same OS is present on all nodes in both cases. 3) As multiple processors present in single machine, no of copies of multiprocessor OS is one. 4) In multiprocessor OS resource management is global and central. 5) Both multiprocessor OS and multicomputer OS are not open. These OS are developed to optimize the performance.	1) Degree of transparency is low. 2) OS on different nodes is different. 3) As multiple machines are present multiple copies of NOS are required. 4) In network OS resource management is per node. 5) Network OS is openness. Modern operating systems are built with microkernel design.	1) Degree of transparency is high. 2) OS on different nodes is different. 3) Middleware is installed on the top of NOS on every machine. 4) Resource management is per node. 5) It is also open.

Q.3 Define Remote Procedure Call. Describe the working of RPC in detail.

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- Program executing on machine A calls procedure or function that is located at machine B is called as RPC.
- Although this definition seems to be simple but involves some technical issues.
- The primitives used in sending and receiving the messages do not hide communication.
- In RPC, calling program calls remote procedure by sending parameters in call and callee returns the result of procedure.
- Working:



Step 1: Client calls the client stub. As client stub is on client machine, this call is a local procedure call, and parameters pushed onto the stack in the normal manner.

Step 2: Client stub packs the parameters into a message and issue a system call to send the message. This process is called marshalling.

Step 3: Kernel sends the message from the client machine to the server machine.

Step 4: Kernel on server machine, send the incoming packet to the server stub.

Step 5: Server stub calls the server procedure.

Q.4 Write a note on Group Communication.

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- In distributed system, it is necessary to have support for sending data to multiple receivers.
- This type of communication is called multicast communication.
- To support multicast communication, many network level and transport-level solutions have been implemented.
- The issues in these solutions were to setup the path to disseminate the information.
- Alternative to these solutions is application level multicasting techniques as peer to peer solutions are usually deployed at application layer.
- It now became easier to setup communication paths.
- Multicast messages offers good solution to build distributed system with following characteristics.

1) Multicast Fault tolerance based on replicated service

⇒ Replicated service is available on group of servers. Client requests are multicast to all these servers in groups. Each server performs the same operation on this request. If some member of group fails, still client will be served by other active members.

2) Finding discovery servers in spontaneous networking
⇒ Servers and clients can use multicast messages to find discovery services to register their interfaces or to search the interfaces of other services in the distributed system.

3) Better performance through replicated data
⇒ Replication improves performance. IF in some cases, replica of data is placed in client machine then changes in data items is multicast to all the processes managing the replicas.

4) Event notification
⇒ Multicast to group to notify if some event occurs.

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